AI511/MM505 Linear Algebra with Applications

Lecture 4 – Inverses (continued)

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Schedule

Week	Lectures	Exercise sessions
36	Theory	No exercise sessions
37	Theory	Theory
38	**Applications**	**Applications**
39	Theory	Theory
40	Theory	Theory
41	Theory	Theory
42	Autumn break	
43	Theory	Theory
44	Applications	Theory

- Week 38 is an introduction to linear algebra in Python.
- Week 44 is split: lectures cover applications, exercises cover theory.

Outline

Recap

Matrices and matrix operations (continued)

- Inverse of a matrix (continued)
- Systems of linear equations revisited

Recap

Systems of linear equations and the inverse of a matrix

- A system of linear equations can be compactly written as

$$Ax = b, \tag{1}$$

  where $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^{n \times 1}$, and $b \in \mathbb{R}^{m \times 1}$.
- Can we somehow “divide” both sides of the equation by A so that we obtain the solution? What kind of a matrix does A have to be so that we could “divide” both sides of the equation?

Inverse of a matrix

- We say $A \in \mathbb{R}^{n \times n}$ is *invertible* if there exists a matrix, which we denote A^{-1} and call the *inverse* of A, such that
$$AA^{-1} = A^{-1}A = I_n.$$
- Note that the definition of the inverse of a matrix is only applicable to square matrices.
- If a matrix is invertible, its inverse is unique.
- Suppose that
$$A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$
Which of these two matrices is invertible? Is there anything peculiar about one of these two matrices?

Questions

- Are all matrices invertible?
- How to check if a general matrix is invertible?
- If a matrix is invertible, how do we find its inverse?

Matrices and matrix operations (continued)

Matrices and matrix operations (continued)

Inverse of a matrix (continued)

Definition

We say that a system of linear equations is *homogeneous* if it can be expressed in matrix form as

$$Ax = 0,$$

where $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^{n \times 1}$.

A homogenous system of linear equations always has at least one solution.

Characterising invertibility

Theorem 2.2.4 of Johnston (2021)

Suppose that $A \in \mathbb{R}^{n \times n}$. The following are equivalent.

(a) A is invertible.

(b) The linear system $Ax = 0$ has a unique solution.

(c) The reduced REF of A is I_n.

- The reduced REF tells us if A^{-1} exists but how do we find it?
- For that we need elementary matrices.

Definition

A square matrix $E \in \mathbb{R}^{n \times n}$ is called an *elementary matrix* if it can be obtained from the identity matrix via a single elementary row operation.

Examples

$$\underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_1 \leftrightarrow R_2}, \quad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}}_{R_3 - 3R_1}, \quad \text{and} \quad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\frac{1}{2}R_2}.$$

Elementary row operations as matrix multiplication

Theorem B.3.1 of Johnston (2021)

Suppose $A \in \mathbb{R}^{m \times n}$. If applying a single elementary row operation to A results in a matrix B, and applying that same elementary row operation to I_m results in a matrix E, then $EA = B$.

For example,

$$\underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{=E} \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}}_{=A} = \underbrace{\begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}}_{=EA}.$$

Gauss-Jordan elimination as matrix multiplication

- Multiplying on the left by an elementary matrix is equivalent to performing a single elementary row operation.
- Multiplying on the left by several elementary matrices, one after another, is equivalent to performing a sequence of elementary row operations.
- This implies that the row operations in Gaussian or Gauss-Jordan elimination can be represented as matrix multiplication.

Finding the inverse of a matrix

- According to Theorem 2.2.4 of Johnston (2021), A is invertible if and only if a finite number of elementary row operations transforms A into I_n or, equivalently, there exist elementary matrices $E_1, \dots, E_k$ such that

$$E_k \dots E_1 A = I_n.$$

- If A is invertible, it follows that

$$E_k \dots E_1 = E_k \dots E_1 I_n = E_k \dots E_1 A A^{-1} = I_n A^{-1} = A^{-1}.$$

- In simple terms, the elementary row operations that bring A to I_n simultaneously bring I_n to A^{-1}.

Computing the inverse of a matrix

Theorem 2.2.5 of Johnston (2021)

Suppose $A \in \mathbb{R}^{n\times n}$. Then A is invertible if and only if there exists a matrix $E \in \mathbb{R}^{n\times n}$ such that the reduced REF of the block matrix $\left[\begin{array}{c|c} A & I_n \end{array}\right]$ is $\left[\begin{array}{c|c} I_n & E \end{array}\right]$. Furthermore, if A is invertible then it is necessarily the case that $A^{-1} = E$.

Exercise

- Suppose that
$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$
- Find the inverse of A by bringing the augmented matrix
$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$
  to a reduced REF.
- If the reduced REF is of the form $[I_3 \mid B]$, then A is invertible and $B = A^{-1}$. If not, then A is not invertible.

A simplifying theorem

Theorem 2.2.7 of Johnston (2021)

Suppose that $A \in \mathbb{R}^{n\times n}$. If there exists a matrix $B \in \mathbb{R}^{n\times n}$ such that $AB = I_n$ or $BA = I_n$, then A is invertible and $A^{-1} = B$.

Proof.

If $BA = I_n$, then multiplying the linear system $Ax = 0$ on the left by B shows that $BAx = B0 = 0$, but also $BAx = I_n x = x$, so $x = 0$. The linear system $Ax = 0$ thus has $x = 0$ as its unique solution, so it follows from Theorem 2.2.4 of Johnston (2021) that A is invertible. Multiplying the equation $BA = I_n$ on the right by A^{-1} shows that $B = A^{-1}$, as desired. If $AB = I_n$, the argument is similar (hint: consider the transpose of AB). The proof is complete. $\square$

Matrices and matrix operations (continued)

Systems of linear equations revisited

Unique solution of a system of linear equations

Part (c) of Theorem 2.2.4 Johnston (2021)

Suppose that $A \in \mathbb{R}^{n \times n}$. The following are equivalent:

(a) A is invertible;

(b) the system of linear equations $Ax = b$ has a unique solution for all $b \in \mathbb{R}^{n \times 1}$.

Proof.

We prove (a) $\implies$ (b). Since $A(A^{-1}b) = b$, it follows that $x = A^{-1}b$ is a solution. Suppose that y is another solution, i.e., $Ay = b$. Then

$$x - y = (A^{-1}A)(x - y) = A^{-1}(A(x - y)) = A^{-1}0 = 0.$$

Hence, $A^{-1}b$ is a unique solution. The proof is complete. $\square$

Solutions of systems of linear equations

Theorem 2.1.1 of Johnston (2021)

Every system of linear equations has either

(a) no solutions;

(b) exactly one solution;

(c) infinitely many solutions.

Proof.

Assume that x_1 and x_2 are two solutions (i.e., $Ax_1 = Ax_2 = b$) such that $x_1 \neq x_2$ and consider another vector $(1-c)x_1 + cx_2$ for $c \in \mathbb{R}$. Then

$$A((1-c)x_1 + cx_2) = (1-c)Ax_1 + cAx_2 = (1-c)b + cb = b$$

which means the system has an infinite number of solutions since $(1-c)x_1 + cx_2$ is a solution for any $c \in \mathbb{R}$. The proof is complete. $\square$

References

Johnston, Nathaniel (2021). ***Introduction to linear and matrix algebra.*** Springer Cham.